

Department of Computer Science and Engineering
Indian Institute of Technology Jodhpur
CSL7490: Introduction to Blockchain (Quiz - 3)

Time: 30 Mins Date: 16 April 2026, Thursday Maximum Marks: 20

Name:

Roll Number:

Please read the following instructions carefully.

- All questions are compulsory. Each question carries one mark.
- Students must write their Name and Roll Number clearly on the question paper.
- Each question must be answered in exactly one sentence only, written directly on the question paper.
- No separate answer sheet shall be provided or accepted under any circumstances.
- Wherever required, students must clearly state and justify any assumptions made in their answers.
- Any form of copying or unfair means will result in zero marks in this quiz and may lead to further disciplinary action, including reduction of final course grades, as per the institute's rules.

1. In the Gasper consensus mechanism, what is the specific requirement for a checkpoint to transition from Justified to Finalized?
- (a) It must be signed by at least 51% of the active hash power.
 - (b) Its direct child checkpoint and justified by a $\frac{2}{3}$ supermajority.
 - (c) It must stay at the head of the chain for exactly 32 slots.
 - (d) The RANDAO value must be revealed by the next proposer.

Answer: (b)

2. What is the consequence of *equivocation* (signing two different blocks for the same slot)?
- (a) A temporary 5-minute suspension from the network.
 - (b) A reduction in the validator's Priority Fee.
 - (c) Immediate slashing, and the validator is forcibly removed.
 - (d) The transaction is replayed on the Beacon Chain.

Answer: (c)

3. In the context of LMD GHOST, what does the Latest Message rule specifically prevent?
- (a) Reentrancy attacks on the Deposit Contract.
 - (b) The use of r, s, v components from previous transactions.
 - (c) The burning of the Base Fee during a hard fork.
 - (d) Exited validators try to influence the current head of the chain.

Answer: (d)

4. A validator has an Effective Balance of 32 ETH. After several months of perfect uptime, their actual balance on the Beacon Chain is 34.2 ETH. Which statement accurately describes their current status?
- (a) Their probability of being a Block Proposer has increased by 6.8%.

- (b) Their voting weight in a committee is now calculated based on 34 ETH.
- (c) 32 ETH is effective balance and the extra 2.2 ETH are non-voting liquidity.
- (d) They are automatically slashed for exceeding the protocol's stake limit.

Answer: (c)

5. Which of the following is a Safety property in a consensus protocol?

- (a) Nothing bad ever happens.
- (b) Something good eventually happens.
- (c) The system responds to all requests within 12 seconds.
- (d) The validator set is reshuffled every epoch.

Answer: (a)

6. The Inactivity Leak is a quadratic penalty mechanism that ensures the network can eventually restore Finality even if more than $\frac{1}{3}$ of the validators are offline. Ethereum's Inactivity Leak is an architectural mechanism specifically designed to trade:

- (a) Safety for Consistency
- (b) Safety for Liveness
- (c) Availability for Gas Efficiency
- (d) RANDAO entropy for Slot duration

Answer: (b)

7. The Inactivity Leak is triggered when the Ethereum network fails to reach finality for more than _____ epochs, causing the protocol to begin penalizing offline validators.

Answer: 4

8. Randomness is a critical component of a permissionless blockchain protocol, serving as a foundation for both _____ and _____ in validator selection.

Answer: Security and Fairness

9. The RANDAO mechanism in Ethereum provides a decentralized way to generate a seed that is intended to be both _____ and _____ for all network participants.

Answer: Verifiable and Deterministic

10. Why is the Ethereum Commit-Reveal based RANDAO scheme described as being Unpredictable and Unstoppable, but only Weakly Unbiasable?

Answer: Weakly Unbiasable: While a validator cannot change the value of their signature, they can choose to withhold it (by not proposing). This allows the last actor in an epoch to choose between two potential future seeds (one with their reveal and one without), providing a 1-bit influence.

11. What is the primary purpose of shuffling validators into different committees every epoch?

- (a) To reduce the amount of ETH required to run a validator node.
- (b) To ensure that every validator becomes a Block Proposer exactly once per epoch.
- (c) To increase the speed of transaction execution on the execution layer.
- (d) To prevent attackers from predicting validator groups for a 51% committee attack.

Answer: (d)

12. Which component is responsible for performing the shuffling calculation locally on each node?

- (a) Consensus Client

- (b) Execution Client
- (c) Validator Client
- (d) The Hardware (The Ethereum Foundation's master server)

Answer: (a)

13. If the RANDAO seed is biased by a Last Revealer Attack, what is the direct impact on shuffling?

- (a) All validators are immediately slashed.
- (b) The network will stop producing blocks entirely.
- (c) The attacker can potentially influence which slots they are assigned.
- (d) The attacker can double the amount of ETH in their account in the future.

Answer: (c)

14. Which property of the shuffling algorithm ensures that every node in the world reaches the same conclusion about who is in which committee?

- (a) Deterministic
- (b) Liveness
- (c) Subjectivity
- (d) Scalability

Answer: (a)

15. What happens if the shuffling results in a committee having fewer than the target number of validators?

- (a) The validator deposit is increased to 64 ETH.
- (b) The number of committees per slot is reduced.
- (c) A random person from X is invited to validate.
- (d) The slot is skipped and no block is produced.

Answer: (b)

16. If an attacker performs a Short-Range Reorg by proposing a block that ignores the previous slot's block, how does this affect the RANDAO?

- (a) The BLS signatures of the previous slot become invalid.
- (b) The RANDAO Reveal from the orphaned block is still added to the canonical mix.
- (c) It results in two RANDAO mixes existing in competing chain heads.
- (d) The attacker is automatically slashed by Execution Layer.

Answer: (c)

17. An attacker controls 34% of the total staked ETH. Which specific consensus property can they effectively stall indefinitely without being slashed?

- (a) RANDAO Determinism
- (b) Validator Activation Queue
- (c) Safety
- (d) Liveness (Block Production)

Answer: (c), Since safety (finality) requires $> 2/3$ (66.6%) of the stake, an attacker with 34% can withhold their votes to prevent any checkpoint from ever reaching a supermajority.

18. In a committee of 512 validators, 400 are online and 112 are offline. Does the committee's aggregate BLS signature still meet the requirement for that specific committee's attestation? (Yes/ No, justify?)

Answer: Yes, because $400/512 \approx 78\%$, which is greater than 66.6% ($2/3$).

19. Why does the consensus client re-execute or verify execution results through its paired execution client on received blocks?

Answer: So the node can confirm that the proposed state change from the execution payload is valid before attesting to or accepting the block.

20. Why does Ethereum divide validators into committees?

Answer: To keep network load manageable, since not every validator needs to attest in every slot, but all active validators still participate across an epoch.